Decision Tree Learning

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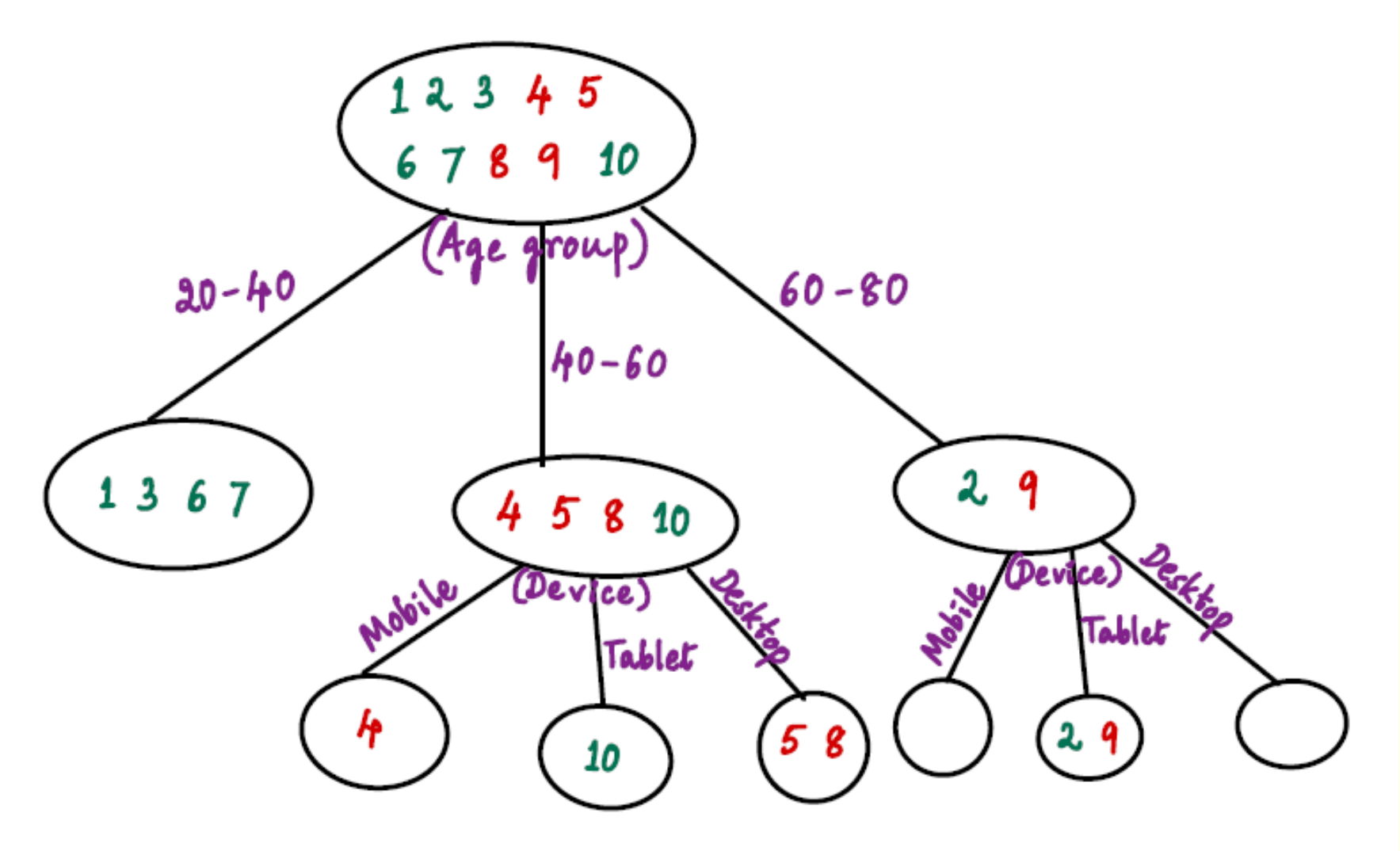
Decision Tree

A Decision Tree is a representation of a function that maps a vector of feature values to a single output value – a *"decision"*.

Consider the following binary classification data from an online shopping site for predicting whether a customer buys a given product or not based on the features "Age group" and "Device used" .

	Age group (years)	Device used	Bought the product
1	20-40	Mobile	Yes
2	60-80	Tablet	Yes
3	20-40	Mobile	Yes
4	40-60	Mobile	No
5	40-60	Desktop	No
6	20-40	Desktop	Yes
7	20-40	Desktop	Yes
8	40-60	Desktop	No
9	60-80	Tablet	No
10	40-60	Tablet	Yes

The following is a decision tree for the above data.

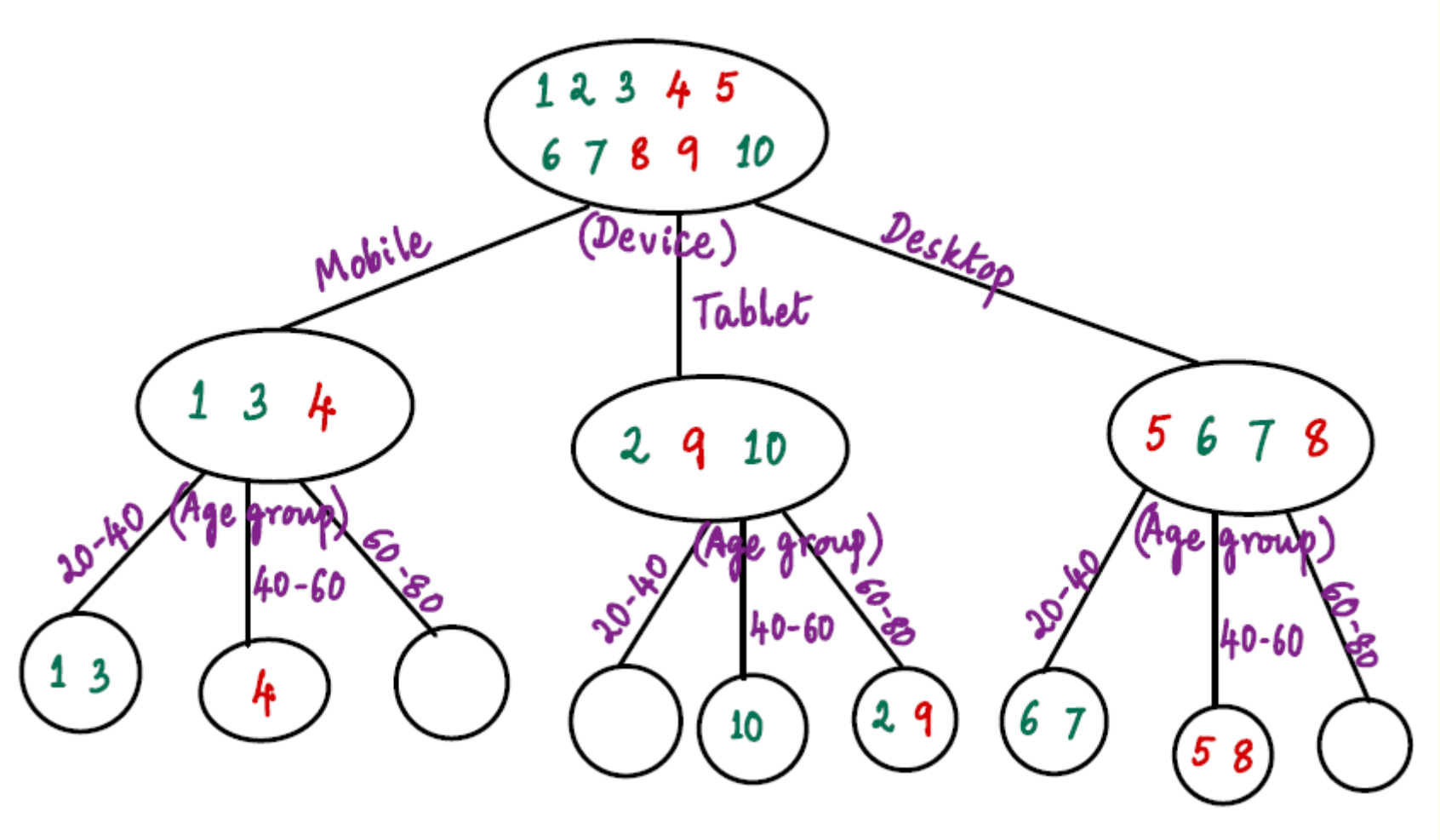


Example prediction:

Age group: 40-60
Device: Tablet

Prediction: Buys the product.

The following is another decision tree for the same data.
Note that the first split is done here based on the feature “Device used”, instead of “Age group” (as in the earlier case).



Example prediction:

Age group: 40-60
Device: Tablet

Prediction: Buys the product.

Decision Tree Learning Algorithm

function LEARN-DECISION-TREE(*examples*, *attributes*, *parent_examples*) **returns** a tree

if *examples* is empty **then return** PLURALITY-VALUE(*parent_examples*)
else if all *examples* have the same classification **then return** the classification
else if *attributes* is empty **then return** PLURALITY-VALUE(*examples*)
else
$A \leftarrow \underset{a \in attributes}{\text{argmax}}\ \text{IMPORTANCE}(a, examples)$

tree ← a new decision tree with root test A
for each value v of A **do**
$exs \leftarrow \{e : e \in examples \text{ and } e.A = v\}$
subtree ← LEARN-DECISION-TREE(*exs*, *attributes* $-A$, *examples*)
add a branch to **tree** with label $(A = v)$ and subtree *subtree*
return *tree*

The function PLURALITY-VALUE selects the most common output value among a set of examples, breaking ties randomly.

At every step, the attribute (feature) to be used for splitting the decision tree nodes is determined by the function IMPORTANCE.

Which is the best feature (attribute) for the first split of the decision tree in the example shown earlier?

The importance of attributes is measured using the notion of **Information Gain**.

It is defined in terms of **Entropy**.

The notion of Entropy (from Information Theory)

The entropy of a random variable quantifies the uncertainty or information associated with its possible outcomes.

Given an event E, the *information content* of E is a function that decreases as the probability $p(E)$ decreases. It is given by

$$I(E) = \log_b \frac{1}{p(E)},$$

or equivalently,

$$I(E) = -\log_b p(E) .$$

Note that when $p(E) = 1$, we get $I(E) = 0$.

Entropy is the expected information content of the random variable, that is,

$$\begin{aligned} H(X) &= \boldsymbol{E}[I(X)] \\ &= \boldsymbol{E}[-\log_b p(X)] \\ &= -\sum_{x\epsilon S} p(x) \log_b p(x). \end{aligned}$$

The choice of base for b for the logarithm varies for different applications. Base 2 is usually used. It gives the unit of bits (or "shannons").

For a Boolean random variable V which is true with a probability q, we have

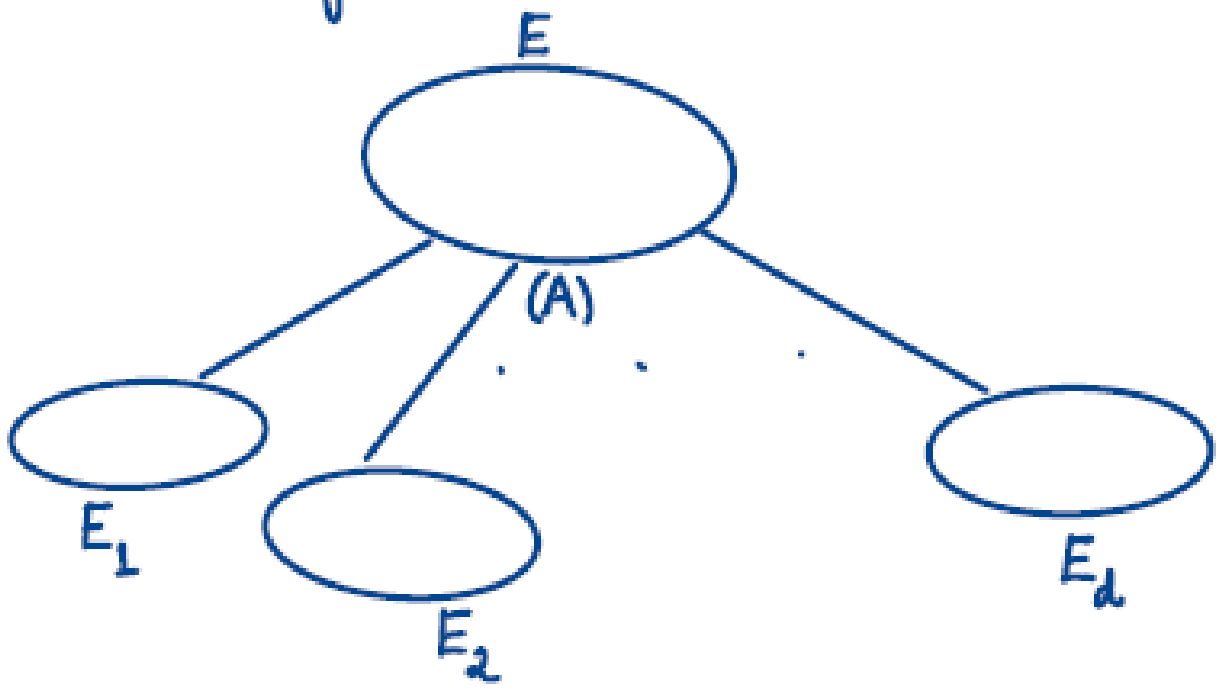
$$H(V) = -(q \log_2 q + (1-q) \log_2 (1-q))$$

Note that this is a function of q alone. It is denoted by $B(q)$.

Note: $B(q) = B(1-q)$.

Let E be a set of examples and let A be an attribute (feature) with d distinct values.

Let $E_1, E_2, \ldots, E_d$ be the subsets of examples formed after testing attribute A.



Let V be the random variable that represents the class to which an example in E belongs.

For each i $(1 \leq i \leq d)$, let V_i be the random variable that represents the class to which an example in E_i belongs.

Entropy in E before testing attribute $A = H(V)$.
(Initial entropy)

Expected entropy after testing A (Remainder(A))

$$= \sum_{i=1}^{d} P(E_i) \cdot H(V_i).$$

Information Gain (A) (Expected reduction in entropy for A)

= Initial entropy − Remainder(A).

Information Gain is a commonly used IMPORTANCE function in the Decision Tree Learning Algorithm.

In a binary classification problem,
if a training set contains p positive examples and
n negative examples, then the entropy of the output
variable on the whole set is $B\left(\frac{p}{p+n}\right)$.

Suppose $E_1, E_2, \ldots, E_d$ are the subsets of examples
formed after testing attribute A. For each i $(1 \leq i \leq d)$,
let p_i and n_i be the number of positive and
negative examples respectively in E_i.

Note that $P(E_i) = \dfrac{p_i + n_i}{p + n}$.

$$\text{Remainder}(A) = \sum_{i=1}^{d} \left(\frac{p_i + n_i}{p + n} \right) \cdot B\left(\frac{p_i}{p_i + n_i} \right).$$

$$\text{Information Gain}(A) = B\left(\frac{p}{p+n} \right) - \text{Remainder}(A).$$

For the binary classification data given earlier, we compute the information gains associated with the two attributes (features), “Age Group” and “Device”:

$$\text{Initial entropy} = B\left(\frac{6}{10}\right)$$

$$= -\left[\frac{6}{10}\log_2\frac{6}{10} + \frac{4}{10}\log_2\frac{4}{10}\right].$$

$$\approx 0.9710 \text{ bits}.$$

$$\text{Remainder}(\text{"Age Group"}) = \frac{4}{10} B\left(\frac{4}{4}\right) + \frac{4}{10} B\left(\frac{1}{4}\right) + \frac{2}{10} B\left(\frac{1}{2}\right)$$

$$= \frac{4}{10}(0) + \frac{4}{10}\left[-\left(\frac{1}{4}\log_2\frac{1}{4} + \frac{3}{4}\log_2\frac{3}{4}\right)\right]$$

$$+ \frac{2}{10}\left[-\left(\frac{1}{2}\log_2\frac{1}{2} + \frac{1}{2}\log\frac{1}{2}\right)\right]$$

$$\approx 0.5245 \text{ bits.}$$

$$\text{Information Gain}(\text{"Age Group"}) \approx 0.9710 - 0.5245$$

$$= 0.4465 \text{ bits.}$$

$$\text{Remainder}(\text{"Device"}) = \frac{3}{10} B\left(\frac{2}{3}\right) + \frac{3}{10} B\left(\frac{2}{3}\right) + \frac{4}{10} B\left(\frac{2}{4}\right)$$

$$= \frac{3}{10}\left[-\left(\frac{2}{3}\log_2\frac{2}{3} + \frac{1}{3}\log_2\frac{1}{3}\right)\right]$$

$$+ \frac{3}{10}\left[-\left(\frac{2}{3}\log_2\frac{2}{3} + \frac{1}{3}\log_2\frac{1}{3}\right)\right]$$

$$+ \frac{4}{10}\left[-\left(\frac{2}{4}\log_2\frac{2}{4} + \frac{2}{4}\log_2\frac{2}{4}\right)\right]$$

$$\approx 0.9510 \text{ bits.}$$

$$\text{Information Gain}(\text{"Device"}) \approx 0.9710 - 0.9510$$

$$= 0.02 \text{ bits.}$$

Since

Information Gain ("Age Group") > Information Gain ("Device"),

"Age Group" is the best feature for the first split of the decision tree.

Consider the following multiclass classification data taken from a restaurant for predicting customer feedback based on the type of cuisine ordered and the dining time.

Cuisine	Dining Time	Customer Feedback
Italian	Evening	Positive
Italian	Afternoon	Positive
Italian	Morning	Neutral
Italian	Evening	Neutral
Italian	Afternoon	Positive
Mexican	Evening	Neutral
Mexican	Afternoon	Negative
Mexican	Morning	Negative
Mexican	Evening	Neutral
Mexican	Afternoon	Neutral
Chinese	Evening	Negative
Chinese	Afternoon	Positive
Chinese	Morning	Negative
Chinese	Evening	Positive
Chinese	Afternoon	Negative
Italian	Morning	Positive
Mexican	Morning	Neutral
Chinese	Morning	Neutral
Italian	Evening	Negative
Mexican	Evening	Positive

Which is the best feature for the first split of the decision tree for the classification task based on the above data?

$$\text{Initial entropy} = -\left[\frac{7}{20}\log_2\frac{7}{20} + \frac{7}{20}\log_2\frac{7}{20} + \frac{6}{20}\log_2\frac{6}{20}\right].$$

$$\approx 1.58129 \text{ bits}.$$

Remainder("Cuisine")

$$= \frac{7}{20}\left[-\left(\frac{4}{7}\log_2\frac{4}{7} + \frac{2}{7}\log_2\frac{2}{7} + \frac{1}{7}\log_2\frac{1}{7}\right)\right]$$

$$+ \frac{7}{20}\left[-\left(\frac{1}{7}\log_2\frac{1}{7} + \frac{4}{7}\log_2\frac{4}{7} + \frac{2}{7}\log_2\frac{2}{7}\right)\right]$$

$$+ \frac{6}{20}\left[-\left(\frac{2}{6}\log_2\frac{2}{6} + \frac{1}{6}\log_2\frac{1}{6} + \frac{3}{6}\log_2\frac{3}{6}\right)\right]$$

$$\approx 1.40289 \text{ bits.}$$

$$\text{Information Gain("Cuisine")} \approx 1.58129 - 1.40289$$

$$= 0.178398 \text{ bits.}$$

Remainder("Dining Time")

$$= \frac{6}{20}\left[-\left(\frac{1}{6}\log_2\frac{1}{6} + \frac{3}{6}\log_2\frac{3}{6} + \frac{2}{6}\log_2\frac{2}{6}\right)\right]$$

$$+ \frac{6}{20}\left[-\left(\frac{3}{6}\log_2\frac{3}{6} + \frac{1}{6}\log_2\frac{1}{6} + \frac{2}{6}\log_2\frac{2}{6}\right)\right]$$

$$+ \frac{8}{20}\left[-\left(\frac{3}{8}\log_2\frac{3}{8} + \frac{3}{8}\log_2\frac{3}{8} + \frac{2}{8}\log_2\frac{2}{8}\right)\right]$$

$$= 1.5 \text{ bits.}$$

Information Gain("Dining Time") $\approx 1.58129 - 1.5$

$= 0.08129$ bits.

Since

Information Gain ("Cuisine") > Information Gain ("Dining Time"),

"Cuisine" is the best feature for the first split of the decision tree.

Decision Trees for Regression (Regression Trees)

What is a Regression Tree?

- A decision tree used for predicting **numeric values.**
- Splits data into regions where targets are similar.
- Each leaf predicts the **mean** of its samples.

How It Works:

- Start with all data at the **root.**

- For each possible split:
  - Choose an attribute and threshold.
  - Partition data into two groups.

- Measure how good the split is using **variance reduction** or **mean squared error (MSE).**

- Select the split with the largest improvement.

- Repeat recursively on each branch.

Stopping & Prediction:

- Stop splitting when one of the conditions is met:
  - Maximum depth is reached.
  - There are a few samples in a leaf.
  - There is no significant variance reduction.

- Leaf prediction = **Mean target value** of samples in the leaf.

Variance at a node S, $Var(S) = \frac{1}{|S|}\sum_i\{(y_i - \bar{y})\}^2$.

If S_L and S_R are the left and right child nodes of a node S, then

$$\text{Variance reduction} = Var(S) - \left(\frac{|S_L|}{|S|}Var(S_L) + \frac{|S_R|}{|S|}Var(S_R)\right).$$

Note:

The variance at a node is the same as the mean squared error (MSE), because the predictions for the samples in a node are the same as the mean target value of the samples in the node.

Hence, in regression trees, maximizing variance reduction is equivalent to minimizing the mean squared error (MSE).